

Changing Drawing Appearance

CADD allows you to change the appearance of drawing objects in a number of ways. You can change the line type, color, size and style of all the drawing objects. Using the editing functions of CADD, you can change almost every factor that defines a drawing object.

When you enter a command to change the drawing appearance, you are prompted to select the objects you want to change. You can select the objects using any of the methods discussed earlier. After the selection is made, you can specify what you want to change, and the selected objects are changed accordingly.

The edit functions allow you to do the following:

- Change line type, color and layer of drawing objects.
- Change the radius value of arcs and circles.
- Change major-axis, minor-axis and rotation angle of the ellipses.
- Change the thickness and flow of splines.
- Change the height, width, spacing, inclination, fonts and justification of text.
- Change the size, style, units, and accuracy of dimensions.
- Change the size and style of symbols, arrows, borders, and patterns.

Using Edit Functions as a Drawing-Aid Tool

The edit functions enable you to manipulate diagrams and make quick corrections in the following manners:

- Cutting drawing objects along an edge
- Drawing parallel lines and arcs
- Extending drawing objects to an edge
- Dividing an object into equal parts
- Making sharp and rounded corners
- Making chamfered corners
- Stretching a diagram
- Enlarging or reducing a diagram
- Rotating a diagram
- Mirroring a diagram

Cutting Drawing Objects Along an Edge

CADD allows you to erase drawing objects along a selected edge (this technique is often called trimming). When you use this function, you are prompted to select the drawing object that is to be used as the cutting edge and then select the objects that are to be erased along that edge.

Fig. 5.5 illustrates how you can erase lines along the edge of an arc. The diagram to the left (original) shows the lines crossing the arc. The second diagram shows that the arc is selected as the cutting edge. In the next diagram, the lines falling outside the arc are selected as the lines to be trimmed. The fourth diagram shows the result of trimming the lines along the arc.

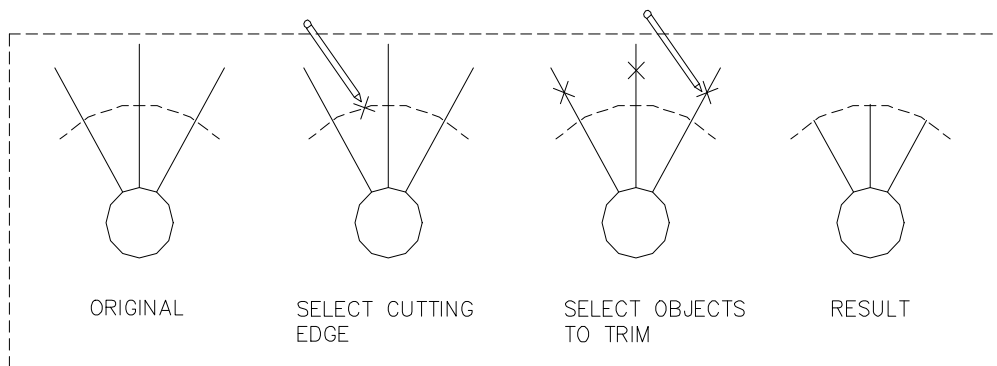


Fig. 5.5: *Trimming lines along an arc.*

Drawing Parallel Lines and Arcs

CADD allows you to draw parallel lines, arcs or splines using previously drawn objects as a reference. This technique is often called offsetting. To draw an offset parallel line, you need to select the drawing object to be used as a reference and then specify the distance and direction of the offset. A parallel drawing object is drawn at the specified distance.

Fig. 5.6 shows how you can draw multiple lines, arcs and splines using a single object as a reference. The dark lines shown in the diagram represent the original drawing objects. The rest of the drawing objects are drawn as an offset of the original. When the first offset is drawn, it can be used as a reference to draw the next offset.

Note: When drawing an offset to the inside of an arc, you cannot enter a greater value for the offset than the radius of the selected arc. Such an arc is not possible to draw and the computer displays an error message.

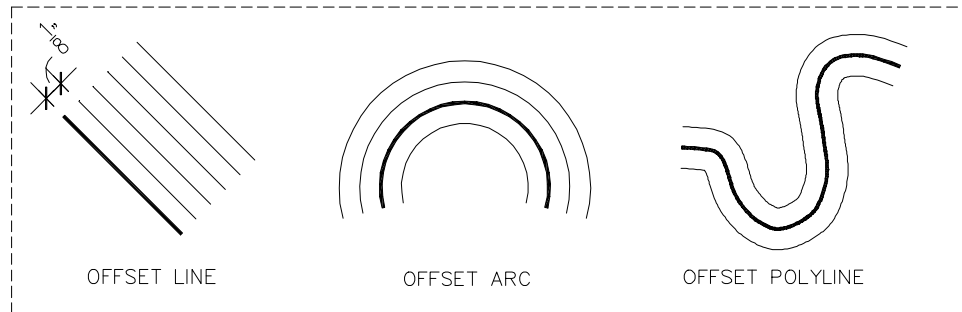


Fig. 5.6: *Drawing offset parallel lines, arcs and splines.*

Extending Drawing Objects to an Edge

CADD allows you to extend lines to a selected drawing object. Often you need to extend lines to construct a drawing and to fix any graphical errors. To extend lines, you need to select an edge to which the lines should extend and then select the lines to be extended. Fig. 5.7 shows how you can extend a number of lines to an edge formed by a line and arcs.

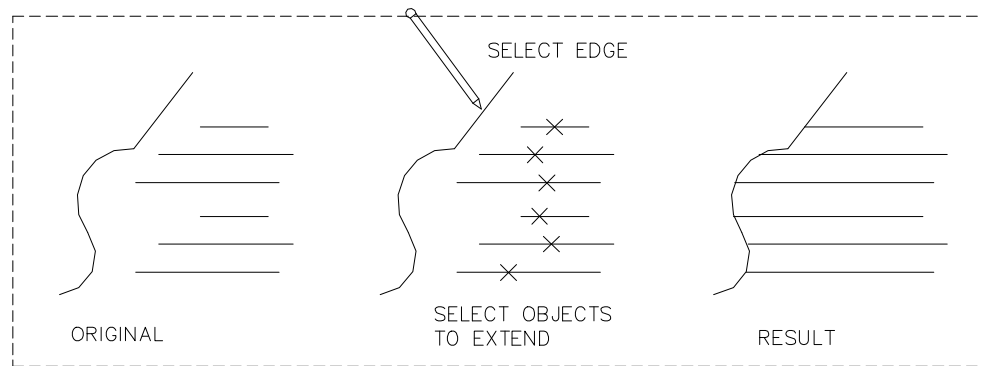


Fig. 5.7: *A number of lines are extended to an edge.*

Dividing an Object into Equal Parts

CADD allows you to place dividing marks on a drawing object such as a line, arc, ellipse or spline. To use this command, you need to select an object and specify how many divisions are required. This function places markers at equal distances on the drawing object.

Fig. 5.8 shows how you can divide an arc, an ellipse, a line and a spline into equal parts. It takes only seconds to place the dividing markers and you don't need to do any calculations. The divisions are accurate to the last fraction and the results can be used for any calculation.

Note: The markers drawn are only for reference, they do not break the drawing object. You can erase the markers when you are finished using them.

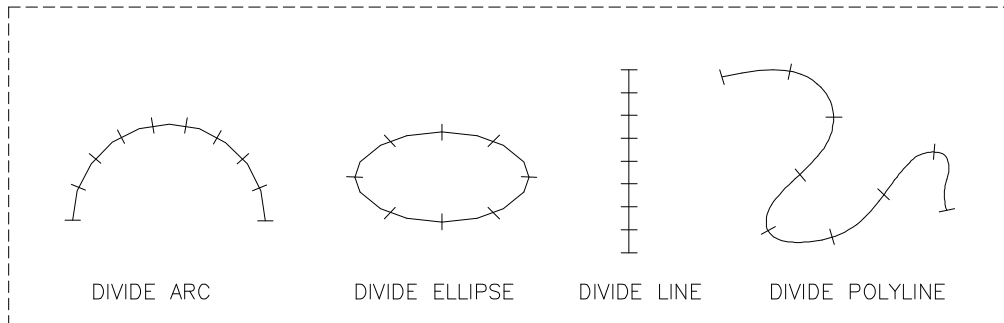


Fig. 5.8: An arc, an ellipse, a line and a spline are divided into equal parts.

Making Chamfered Corners

CADD allows you to make a chamfered corner between two lines. It works quite like the fillet command. When you enter the chamfer command, you are prompted to select the lines that are to be chamfered and enter a chamfer distance. The chamfer distance determines the size of the chamfer. Fig. 5.9 shows how you can make a chamfered corner between two lines that are set apart.

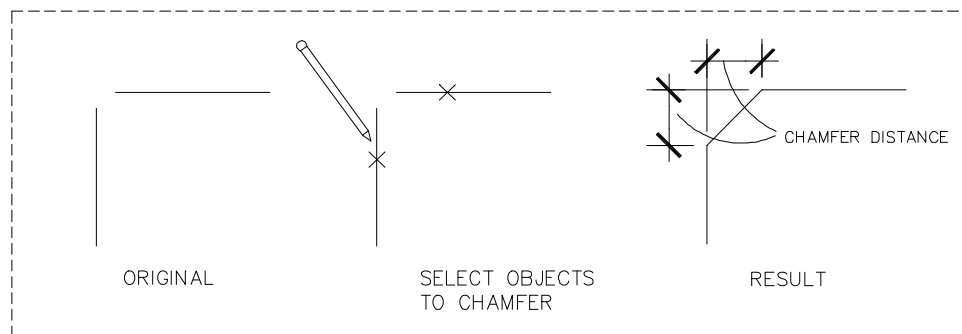


Fig. 5.9: A chamfered corner is made between two lines.

Making Sharp and Rounded Corners

CADD allows you to make fine corners of any two lines or arcs. This technique, often called filleting, is the quickest way to join the missing corners of lines and arcs. With this function active, to make a corner all you need to do is select the lines or arcs that have missing corners. CADD automatically extends or shortens the selected objects to form a corner. You can also specify whether you want a sharp corner or a rounded corner.

Fig. 5.10 shows how you can join the missing corners of lines and arcs. Diagram A shows how to make a sharp corner of two lines that are set apart. Diagram B shows how to make rounded corners between lines and arcs. When making a rounded corner you can specify the radius of the arc that is to be used to create the fillet.

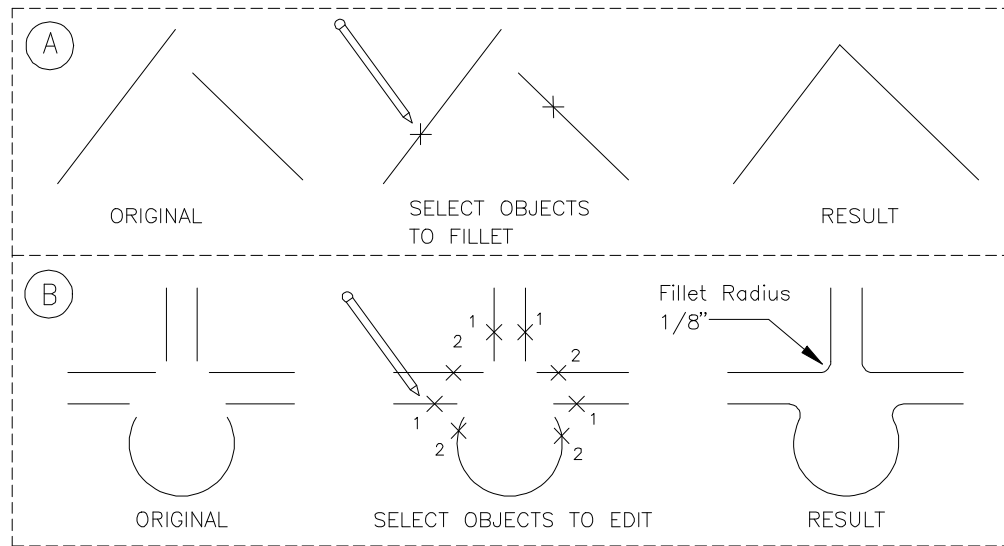


Fig. 5.10: Making sharp and rounded corners of lines and arcs.

Stretching Diagrams

CADD allows you to quickly change the size of diagrams by stretching lines, arcs, splines, etc. This function is very helpful to make quick alterations to drawings. To use the stretch function, you need to select the drawing objects to be stretched and specify the distance and direction of stretching.

Fig. 5.11 shows how you can stretch a rectangular-shaped diagram to make it longer. The objects to be stretched are selected by enclosing them in an editing window formed by points 1 and 2. The diagram is stretched in the horizontal direction by 1".

Note: When dimensions are stretched, the program updates the dimension values automatically. When text and symbols are stretched, their appearance remains unchanged but they are relocated. When only one end of an arc is stretched, it may deform the shape of the arc.

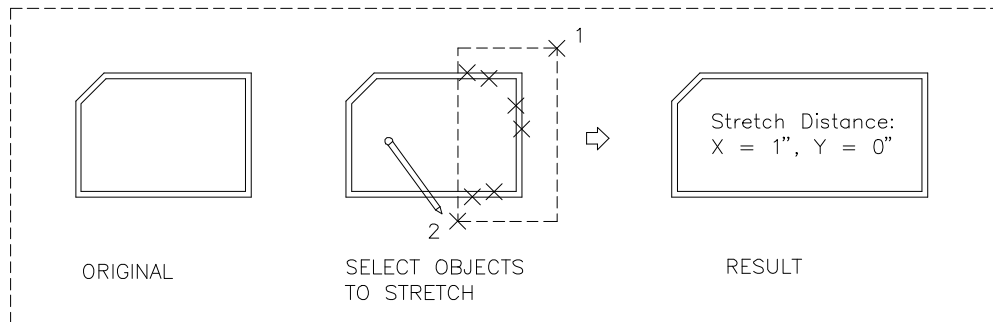


Fig. 5.11: A diagram is stretched in the horizontal direction.

Enlarging or Reducing Diagrams

CADD allows you to enlarge or reduce diagrams in a convenient manner. To enlarge or reduce diagrams, you need to select the objects and enter a scale factor. The scale factor determines by how much the diagrams are to be reduced or enlarged.

Example: To reduce a 10'x20' square to a 5'x10' square, you would need to enter a scale factor of 0.5. Similarly, to enlarge the diagram to twice its original size, you would need to enter a scale factor of 2. You can precisely reduce or enlarge the diagrams by entering an exact scale factor. Fig. 5.12 shows how you might enlarge a portion of a diagram four times to show it in more detail.

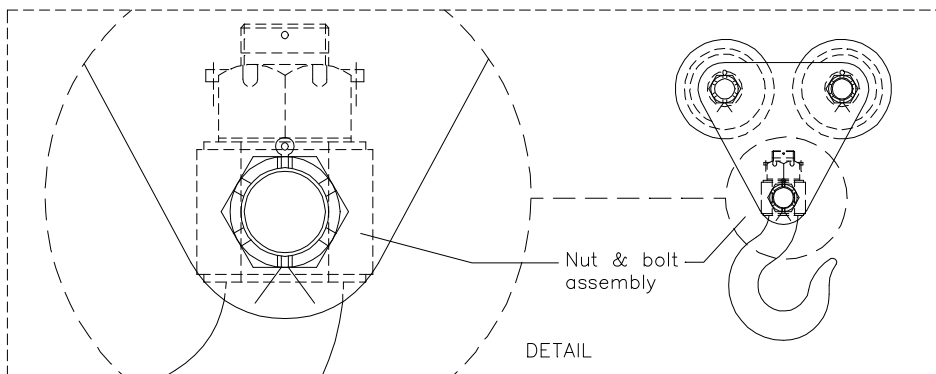


Fig. 5.12: A part of the diagram is enlarged to four times the original size.

Rotating the Diagrams

CADD allows you to rotate selected drawing objects to a specified angle. To rotate, you need to select the drawing objects, enter a reference point (or base point) and the rotation angle. The base point acts as a pivot point around which the objects are rotated. The rotation angle determines by how much the objects will be rotated and in which direction.

Fig. 5.13 illustrates the ways you can rotate a drawing 45° by indicating different base points. The diagram on the left shows how the drawing is rotated when the base point is entered in the middle of the drawing. The diagram in the middle shows the rotation when the base point is entered at a corner. The diagram on the right shows what happens when the base point is entered away from the drawing.

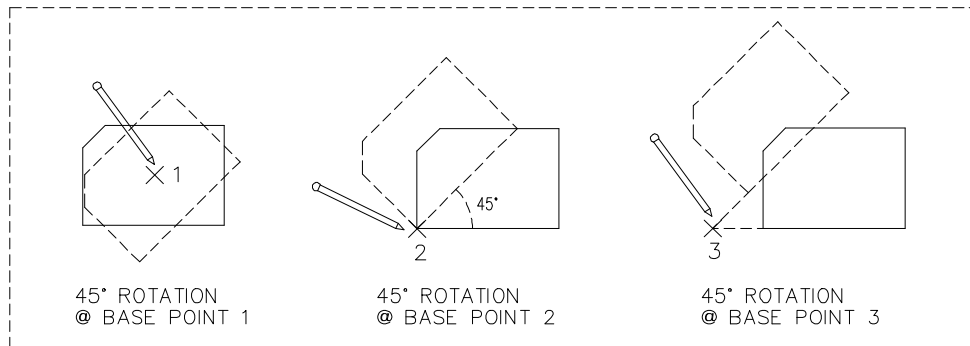


Fig. 5.13: A diagram is rotated at 45°.

Mirroring the Diagrams

CADD allows you to create mirror images of selected drawing objects. This capability is very useful when you want to draw something that is symmetrical on both sides. You need to draw only one half of the drawing; the rest of the drawing can be completed using the mirror function. To make a mirror image, you need to select the objects to be mirrored and indicate a mirror axis. The mirror axis is an imaginary line along which the diagram is mirrored. Fig. 5.14 illustrates how you can draw a complete flange drawing from only half the drawing.

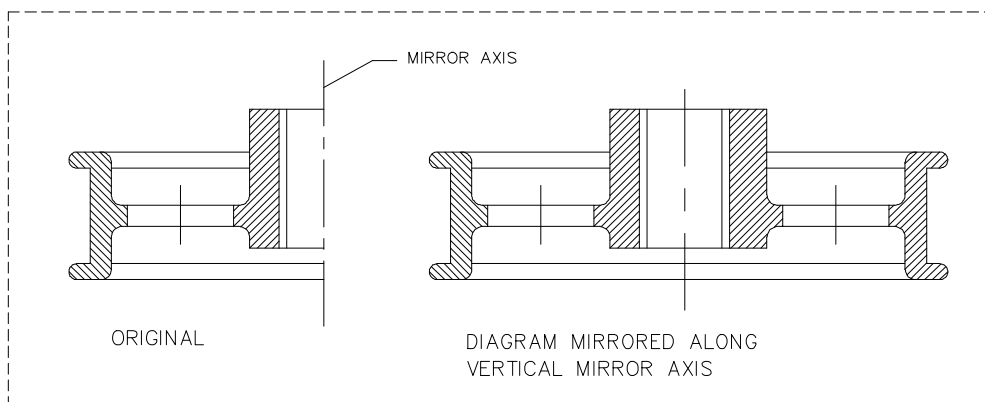


Fig. 5.14: Half of the flange drawing is mirrored to make a complete flange diagram.